Modelling price forecast errors in the RQ3 experiments

The current synthetic model, a renewable-driven alternative, and how they compare

Working note — Carlos Canoyra Calero

25 October 2026

Abstract

RQ3 asks how much welfare is lost when a battery builds its interval bids from an imperfect price forecast instead of the true day-ahead prices. This note documents the error model currently used in the experiments (an additive, parametric, synthetic error on the price), proposes an alternative in which price errors are generated by errors in the renewable forecast and passed through a market model (a structural error), and lists the advantages and drawbacks of both. The recommendation is to keep the synthetic model as the backbone of stages 1–2 and to add the renewable-driven model as an additional error family in stage 3, next to the calibration on historical price forecasts.

Contents

1 Context and the question behind this note **1**

2 The current model: a synthetic, parametric price error **2**
2.1 Formulation . . . 2
2.2 Meaning of the four parameters . . . 3
2.3 How it is used in the experiments . . . 3
2.4 What the pilot used and found . . . 4
2.5 Calibration already planned (stage 3) . . . 4

3 Proposal: price errors driven by renewable forecast errors **4**
3.1 Idea . . . 4
3.2 Formulation . . . 4
3.3 Requirements . . . 5
3.4 Experiments it enables . . . 6

4 Advantages and drawbacks **6**
4.1 Synthetic price error (current model) . . . 6
4.2 Renewable-driven price error (proposal) . . . 7
4.3 Side-by-side summary . . . 8

5 Recommendation **9**

6 Beyond this note: uncertainty after the day-ahead market **9**

1 Context and the question behind this note

In the experiments every battery e receives a forecast $\hat{\lambda}_{e,t}$ of the day-ahead price at its bus, solves its own profit-maximisation (self-scheduling) problem at those prices, selects the multipliers with

the pre-limit rule, builds the equivalent bids

$$u_{e,t} = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e, \qquad c_{e,t} = \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}), \tag{1}$$

and submits them with the quantity rule. The operator clears the current design with the true system, real-time deviations are settled with a premium ψ, and the welfare loss ΔSW is measured against Baldick's benchmark [4] and normalised by the value storage creates in that benchmark. The only imperfect input is $\hat{\lambda}$. *How* $\hat{\lambda}$ is generated is therefore a central modelling choice.

Does the price error already contain the renewable error? For a stand-alone battery in the day-ahead market, largely yes. Everything the battery does not know (wind, solar, demand, outages, fuel prices, flows on interconnectors, other participants' bids) reaches it only through the price. A price error calibrated on real forecasts therefore already contains the renewable forecast error; adding a renewable error *on top of* such a price error would count the same uncertainty twice. Renewable errors add something in three situations only:

1. **The shape of the error.** A generic model (Gaussian, AR(1), the same size in every hour) does not reproduce the structure renewable errors impose on prices: larger errors in hours with a lot of wind or sun, larger and asymmetric errors where net demand sits near a step of the supply curve, and errors shared by all batteries because they all forecast the same weather. Generating price errors *through* renewables gives them that structure. This is the proposal of Section 3.
2. **After the day-ahead market has cleared.** Actual renewable output differs from the day-ahead forecast. That changes the real-time stage for both designs and makes Baldick's own outcome uncertain. It is a second, separate uncertainty that a day-ahead price forecast does not cover (a stochastic extension; out of scope here, see Section 6).
3. **Hybrid plants.** If the battery sits behind a wind or solar plant, its own production error enters its quantities directly, not through prices (this is the setting of Martinez-Rico et al. [1]).

A note on the literature. Martinez-Rico et al. [1] study a wind–solar plant with a battery in the Iberian day-ahead and intraday markets. They perturb *production* forecasts (through wind speed, irradiance and temperature) with a random-walk error whose standard deviation grows as $k\sqrt{\Delta t}$ with the horizon, and test 0–20% at 24 h. Prices, however, are assumed perfect ("no error is applied to the real price"), chosen to isolate the production error. Their conclusion that uncertainty barely affects the benefits refers only to production uncertainty; they note themselves that battery usage is driven by the price spread, which they keep fixed. This is a useful reference for the gap RQ3 addresses: price errors are what matter for storage arbitrage.

2 The current model: a synthetic, parametric price error

2.1 Formulation

Let $\lambda^\star_t$ be the benchmark (Baldick) price at the battery's bus and $\bar{\lambda}$ its mean over the horizon. The forecast of battery e is

$$\hat{\lambda}_{e,t} = \lambda^\star_t + \underbrace{b\,\bar{\lambda}}_{\text{bias}} + \underbrace{\sigma\,\bar{\lambda}\,z_{e,t}}_{\text{noise}}, \qquad t = 1, \dots, T, \tag{2}$$

where the standardised error $z_{e,t}$ mixes a component common to all batteries and a battery-specific one,

$$z_{e,t} = \sqrt{\omega}\,\chi_t + \sqrt{1-\omega}\,\eta_{e,t}, \tag{3}$$

and each component follows a stationary AR(1) process with unit variance,

$$x_t = \rho\, x_{t-1} + \sqrt{1-\rho^2}\, w_t, \qquad x_1 \sim \mathcal{N}(0,1), \quad w_t \sim \mathcal{N}(0,1) \text{ i.i.d.}, \qquad x \in \{\chi, \eta_e\}. \tag{4}$$

By construction $\mathbb{E}[z_{e,t}] = 0$, $\mathrm{Var}(z_{e,t}) = 1$, $\mathrm{Corr}(z_{e,t}, z_{e,t-1}) = \rho$ and $\mathrm{Corr}(z_{e,t}, z_{e',t}) = \omega$ for $e \neq e'$. Hence the hourly error $\hat{\lambda}_{e,t} - \lambda^\star_t$ has mean $b\bar{\lambda}$ and standard deviation $\sigma\bar{\lambda}$ in every hour.

2.2 Meaning of the four parameters

Parameter	Controls	Interpretation	Planned values
σ	size	standard deviation of the hourly error, as a share of the mean price	0–40%
ρ	persistence	$\rho = 0$: hour-to-hour noise, errors in the *shape* of the day; $\rho = 0.9$: the whole day shifts, errors in the *level*	0, 0.5, 0.9
b	bias	systematic over- or under-estimation of all prices	-10, 0, $+10$%
ω	common share	0: each battery has its own forecast error; 1: all batteries share the same error (same forecast vendor, same weather model)	0, 0.5, 1

Figure 1 shows forecasts generated with $\sigma = 10\%$ for three settings. With the same average error size, ρ and b change what the battery perceives: noise mostly moves *when* it trades; persistent errors and bias move *how much stored energy seems worth* (π), which shifts the bids of whole segments of the day.

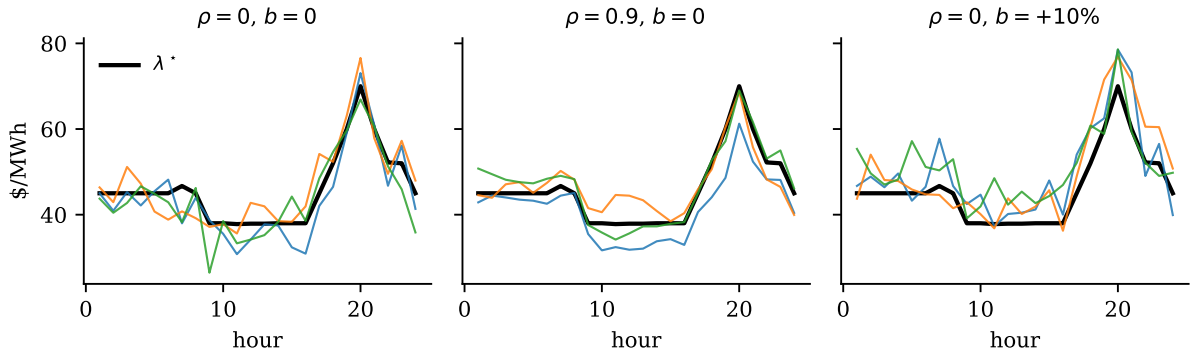


Figure 1: Synthetic forecasts with $\sigma = 10\%$ of the mean price on the pilot day (black: benchmark price $\lambda^\star$; colours: three draws).

2.3 How it is used in the experiments

1. Solve Baldick's benchmark once: $\lambda^\star$, schedule, welfare $SW^\star$.
2. Draw z from (3)–(4) and build $\hat{\lambda}_e$ from (2) for every battery.
3. Each battery solves its own problem at $\hat{\lambda}_e$, applies the pre-limit rule, builds (E) and the quantity rule.
4. The operator clears the current design with the *true* system.
5. Real time: the battery delivers the closest feasible schedule; other units re-dispatch at offer plus ψ; unserved energy at 1000 \$/MWh.
6. Record ΔSW (as % of the storage value), undeliverable energy and price distortion.

Steps 2–6 are repeated for each Monte Carlo draw (150 in the pilot, 500 planned). The four bid families bracket the results: $\mathcal{B}^{\star}$ (true prices and Baldick's quantities, zero loss), $\mathcal{B}^{\star p}$ (true prices, own quantities), $\mathcal{B}^{\text{fcst}}$ (forecast) and $\mathcal{B}^{\text{priv}}$ (technical data only).

2.4 What the pilot used and found

The pilot (24-hour single-bus day, one 100 MW/400 MWh battery, $\bar{\lambda} = 45.9$ \$/MWh, storage value $V \approx 7{,}867$ \$/day, $\psi = 10$, 150 draws) used $b = 0$, a single battery (so ω plays no role), $\rho \in \{0, 0.9\}$ and $\sigma \in \{1, 2, 5, 10, 20\}$ \$/MWh, i.e. 2.2, 4.4, 10.9, 21.8 and 43.6% of the mean price.

σ (% of mean price)	0	2.2	4.4	10.9	21.8	43.6
Loss, $\rho = 0$ (% of V)	11.1	36.5	56.9	65.6	89.1	158.4
Loss, $\rho = 0.9$ (% of V)	11.1	68.0	64.1	102.3	177.2	284.9

The 11.1% at $\sigma = 0$ is the quantity-information loss: with true prices the battery's own plan differs from the schedule the system needs. Technical-data-only bids lose 74.5, 175.4 and 579.0% at $\psi = 0$, 10 and 50. These numbers are indicative (synthetic day with flat price stretches and a stepped supply curve).

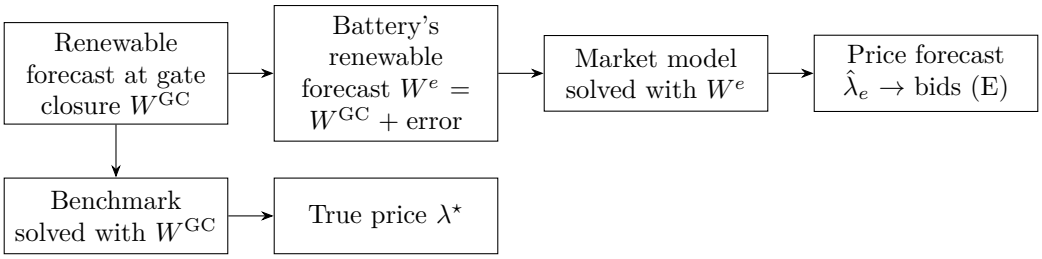
2.5 Calibration already planned (stage 3)

The synthetic grid is meant to be anchored to reality in stage 3 in two ways: (i) estimate (σ, ρ, b) from the errors of a standard day-ahead price forecast (e.g. the LEAR or DNN models of the open `epftoolbox` benchmark [2], run on Danish DK1 prices) and mark that point on the loss curves; (ii) complement the parametric draws with resampled real error days (block bootstrap of historical forecast errors), which keeps their true shape. For roughly normal errors, mean absolute error $\approx 0.8\,\sigma$, so published accuracy figures translate directly into a value of σ.

3 Proposal: price errors driven by renewable forecast errors

3.1 Idea

Instead of perturbing the price directly, perturb the *renewable forecast* the battery uses, and let a market model turn it into a price forecast. The price error is then a consequence of a physical error, and its size, timing, asymmetry and common component follow from the system rather than from assumptions.



3.2 Formulation

Which renewable quantity is uncertain. The battery does not need to predict *actual* renewable output; it needs to predict the *day-ahead price*, which is set by the renewable forecasts that producers submit at gate closure. Let $W^{\text{GC}}_{r,t}$ be the forecast of renewable source r (wind, solar) available to the market at gate closure; it enters the benchmark and defines $\lambda^\star$. The battery forms its view earlier or with a worse model:

$$W^e_{r,t} = \Big[W^{\text{GC}}_{r,t} + \bar{R}_r \big(b_W + \sigma_W \, s_{r,t} \, \xi_{e,r,t}\big) \Big]_0^{\bar{R}_r}, \tag{5}$$

where $\bar{R}_r$ is the installed capacity, b_W and σ_W the bias and size of the renewable error as a share of capacity, $s_{r,t} \in [0,1]$ a profile that makes the error proportional to the possible output (e.g. $s_{r,t} = 0$ at night for solar), $[\cdot]_0^{\bar{R}}$ the clipping to physical limits, and ξ built exactly like z in (3)–(4) with its own (ρ_W, ω_W). A random walk with variance growing with the horizon, as in [1], is a possible alternative for ξ, but in the day-ahead market all hours are forecast at the same moment (horizon about 12–36 h), so a stationary AR(1) with high ρ_W is usually sufficient.

From renewables to prices. The battery's price forecast is the price of the benchmark solved with its own renewable view,

$$\hat{\lambda}_e = \lambda^{\mathrm{SOC}}(W^e), \qquad \lambda^\star = \lambda^{\mathrm{SOC}}(W^{\mathrm{GC}}). \tag{6}$$

This is the price the battery would expect if it knew the market and only misjudged renewables. It is consistent with the rest of the design: with zero renewable error $\hat{\lambda}_e = \lambda^\star$, exactly as $\sigma = 0$ in the synthetic model. Demand errors can be added in the same way.

Optional residual (hybrid model). Renewables are not the only source of price error. A residual synthetic term covers the rest (demand, fuel, outages, other bids):

$$\hat{\lambda}_e = \lambda^{\mathrm{SOC}}(W^e) + \sigma_{\mathrm{res}}\,\bar{\lambda}\,z_{e,t}. \tag{7}$$

With $\sigma_{\mathrm{res}} = 0$ the model is purely structural; with $\sigma_W = 0$ it collapses to the current synthetic model.

Mapping back to the synthetic grid. For every structural setting, compute the statistics of the induced price error $\hat{\lambda} - \lambda^\star$: its size $\hat{\sigma}$ (relative to $\bar{\lambda}$), lag-one autocorrelation $\hat{\rho}$, mean $\hat{b}$, and the cross-battery correlation $\hat{\omega}$. This places each structural run as a point on the synthetic loss curves and tests whether the synthetic model predicts the structural loss. If the loss of a structural run differs from the synthetic loss at the same $(\hat{\sigma}, \hat{\rho}, \hat{b}, \hat{\omega})$, the difference measures how much the *shape* of the error matters.

3.3 Requirements

1. **A smooth supply curve.** The pilot system has ten 110 MW thermal blocks: a staircase. Figure 2 shows the consequence with an illustrative solar error (6% of capacity, $\rho_W = 0.8$, storage ignored): on the staircase the price error is exactly zero in most daylight hours and jumps by a whole block price in a few, so the loss would reflect where the steps are placed rather than how hard forecasting is. A supply curve with many blocks, or a linear marginal cost (quadratic cost), gives a continuous response. Ideally it is calibrated to a real merit order (e.g. DK1).

2. **Two renewable forecasts, not forecast versus actual.** The right comparison is the battery's forecast against the gate-closure forecast. Public data (ENTSO-E Transparency Platform, Energinet) give day-ahead wind and solar forecasts and actual output; the error of the day-ahead forecast against actual output is easy to compute but *overstates* the relevant error, so it should be treated as an upper bound for σ_W.

3. **A renewable share that can be varied.** Scale $\bar{R}_r$ relative to demand (for instance 20, 40 and 60% of energy) so that the link "more renewables ⇒ larger price errors ⇒ larger loss" can be measured.

4. **Consistency of the forecast model with several batteries.** With many batteries, (6) assumes each one anticipates the efficient behaviour of the whole fleet. This is consistent with the price-taking assumption, but should be stated.

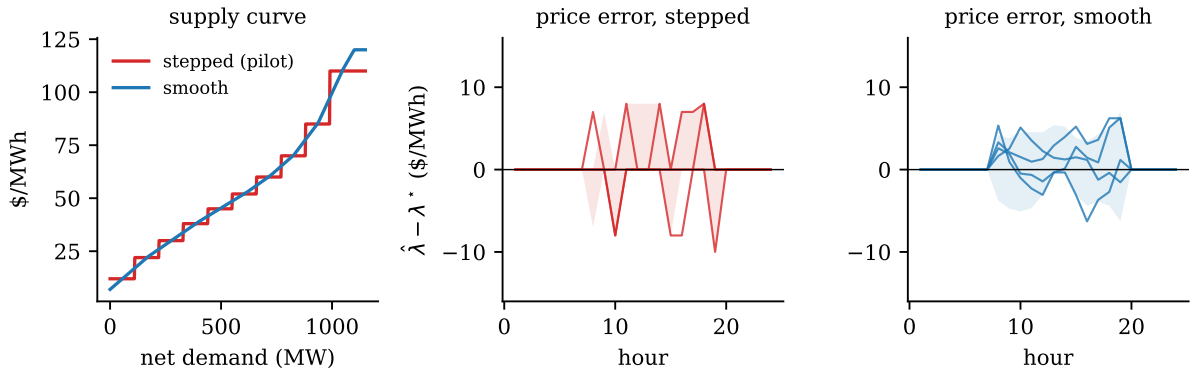


Figure 2: Illustration of a renewable-driven price error on the pilot day without storage. Left: stepped supply curve of the pilot and a smooth one through the same block prices. Centre and right: price error caused by the same solar forecast errors (four draws and 5–95% band). On the staircase most hours have no price error and a few have a full step; on the smooth curve the error is continuous and concentrated in daylight hours.

3.4 Experiments it enables

Axis	Values (indicative)
Renewable share	20, 40, 60% of energy (wind and solar mix)
Renewable error size σ_W	0 to the level of a day-ahead forecast against actual output (upper bound), in 4–5 steps
Persistence ρ_W, bias b_W	as in the synthetic grid
Common share ω_W	1 by default (one weather forecast for all), 0.5 as sensitivity
Residual σ_{res}	0 and one calibrated value
Fleet size	1, 5, 20 batteries ("more renewables, more batteries")

The headline output would be loss as a function of renewable share and renewable forecast quality, plus the induced $(\hat{\sigma}, \hat{\rho}, \hat{b}, \hat{\omega})$ of each run placed on the synthetic map. Computational cost: one extra linear program per draw (if all batteries share the forecast) or per battery and draw.

4 Advantages and drawbacks

4.1 Synthetic price error (current model)

Advantages

1. **Interpretable.** Each parameter isolates one mechanism (size, persistence, bias, common share), so results can be read as "the loss caused by *this* property of the error".
2. **Full control.** The parameters can be set independently and swept on a regular grid. This gives clean loss curves, monotone comparisons and thresholds such as the error level $\sigma^\star$ where a forecast stops beating technical-data-only bids.
3. **General.** The error does not depend on the test system, so conclusions transfer across systems and days, and the normalisation by storage value makes them comparable.
4. **Answers RQ3 as posed.** RQ3 asks what a price error costs, whatever its cause. The synthetic model measures exactly the map from price error to welfare loss, without mixing in a second map (from renewables to prices).

5. **Standard and defensible.** Additive Gaussian noise on prices is common in the literature and easy to explain to an examiner; real forecast errors and their properties are reviewed in [3, 2].
6. **Calibratable.** (σ, ρ, b) can be estimated from historical price forecasts, and resampled real error days can be added to keep the true shape (stage 3).
7. **Common errors already covered.** ω captures shared forecasts across batteries.
8. **Cheap and reproducible.** No extra optimisation per draw; seeds fully determine the draws.
9. **Already built and piloted.** The pipeline, the bid families, the decomposition and the experiment plan are designed around it; no delay to stage 1.

Drawbacks

1. **Assumed shape.** The same standard deviation in every hour, symmetric Gaussian, stationary AR(1). Real price errors are larger in hours with high renewable output and near steps of the supply curve, skewed, heavy-tailed and with occasional spikes.
2. **No causal link to renewables.** The error is independent of the state of the system, so the motivation of the thesis (more renewables, more batteries) is not quantified: the model cannot say how the loss changes with renewable penetration.
3. **Parameters not physical.** "$\sigma = 10\%$" has no direct meaning in terms of wind or solar forecast quality, which makes the realistic range harder to justify without separate calibration.
4. **Forecasts may be implausible.** $\hat{\lambda}$ is not the equilibrium price of any plausible system; with large σ it can be negative or have shapes no forecaster would produce (no clipping is applied).
5. ω **chosen, not derived.** The common share is a free parameter; in reality it follows from how many independent forecasts exist.
6. **Calibration mismatch.** Historical forecast errors come from a real market, while the experiments run on a stylised test system; matching the two (relative to the mean price) is an approximation.

4.2 Renewable-driven price error (proposal)

Advantages

1. **Linked to the motivation.** It measures directly how renewable penetration and renewable forecast quality translate into welfare loss through storage bids.
2. **Physically consistent forecasts.** $\hat{\lambda}_e$ is the equilibrium price of a plausible system, so it never has impossible shapes and stays within the range of the supply curve.
3. **Realistic structure emerges.** Larger errors in high-renewable hours, no error at night for solar, asymmetry near supply-curve steps and nonlinear effects follow from the system instead of being assumed.
4. **Common errors emerge naturally.** If batteries use the same weather forecast, their price errors are correlated; ω becomes meaningful rather than chosen.
5. **Physical parameters.** σ_W is a share of installed capacity and can be calibrated on public renewable forecast data, which is easier to obtain than private price forecasts.
6. **Validates the synthetic grid.** The induced $(\hat{\sigma}, \hat{\rho}, \hat{b}, \hat{\omega})$ places each run on the synthetic curves and shows whether the shape of the error matters beyond its summary statistics.

7. **Natural extensions.** Demand errors fit the same framework, and the operator's own forecast (for reference bids used in market monitoring) can be modelled the same way.

Drawbacks

1. **Strong dependence on the test system.** The induced price error is only as realistic as the supply curve. On the pilot's staircase it produces artefacts (Figure 2); a smooth, calibrated merit order is needed, which is extra modelling with its own assumptions.
2. **Optimistic forecaster.** The battery knows the true market model and every other bid and errs only on renewables. Real price errors also come from demand, fuel and CO_2 prices, outages, interconnector flows and strategic bids; ignoring them underestimates the error.
3. **The residual brings back assumptions.** Covering the other sources needs the synthetic residual (7), so the structural model does not remove parametric assumptions; it adds a layer.
4. **The right data are scarce.** The relevant error is the battery's forecast against the gate-closure forecast; data for that are rarely public. The usual proxy (day-ahead forecast against actual output) overstates it.
5. **Less control.** Size, persistence and asymmetry of the price error come out jointly and depend on the hour, so clean one-parameter sweeps and thresholds are harder to obtain.
6. **Two mechanisms mixed.** The loss combines the renewable-to-price map (a property of the test system) and the price-to-welfare map (the object of RQ3). Separating them requires the mapping exercise above.
7. **Single-bus limitation.** Without a network, local price errors caused by congestion are absent; with a network, the structural model needs the full network in the forecast, which increases cost.
8. **Time and scope.** It requires a new test system and data work, delays stage 1 if done first, and needs the supervisor's agreement on the experiment design.
9. **More computation.** One extra linear program per draw (or per battery and draw); modest, but not zero.

4.3 Side-by-side summary

	Synthetic (current)	Renewable-driven (proposal)
Question answered	cost of a price error of given size, persistence, bias, common share	cost of a renewable forecast error at a given renewable share
Shape of the error	assumed (Gaussian, AR(1), same size every hour)	emerges from the system (hour-dependent, asymmetric)
Parameters	statistical, set freely	physical, set jointly with the system
Dependence on test system	none	strong (supply curve)
Realism of the forecaster	neutral	optimistic unless a residual is added
Link to the motivation	indirect	direct
Data for calibration	historical price forecasts	renewable forecasts (gate-closure timing is hard)
Cost and readiness	cheap, already built	new system and data, extra LPs

5 Recommendation

Keep the synthetic model as the backbone; add the renewable-driven model as an extra family

1. **Stages 1–2: unchanged.** Run the synthetic grid as planned. It answers RQ3 in general and produces the interpretable curves and thresholds.
2. **Stage 3: two ways of grounding the grid in reality.** (a) Calibrate (σ, ρ, b) on historical DK1 price forecast errors and add resampled real error days. (b) Add the renewable-driven model (5)–(7) on a test system with a smooth, calibrated supply curve, varying the renewable share and the renewable error size.
3. **Report the structural runs on the synthetic map.** For each run give the induced $(\hat{\sigma}, \hat{\rho}, \hat{b}, \hat{\omega})$ and compare the loss with the synthetic loss at the same statistics. Agreement validates the synthetic grid; disagreement measures how much the shape of the error matters.
4. **State the limits.** The structural forecaster is optimistic (only renewables uncertain); the renewable error against actual output is an upper bound for the relevant gate-closure error.

Points to agree with the supervisor. (i) Whether the structural family belongs in the thesis or in future work; (ii) the test system for it (single bus with smooth merit order, or a small network); (iii) the data source for σ_W (ENTSO-E or Energinet forecasts against actual output, as an upper bound); (iv) whether to include demand errors in the same framework.

6 Beyond this note: uncertainty after the day-ahead market

Both models above concern what the battery believes *before* the day-ahead market. A different extension is uncertainty *after* it: actual renewable output deviates from the gate-closure forecast, so the real-time stage changes for both designs and Baldick's benchmark itself becomes stochastic. Comparing the two designs then requires a stochastic formulation (scenarios of real-time renewables, expected welfare). This is a natural continuation of the thesis, not part of RQ3.

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